Task 4: AI TOOL EVALUATION

Context

This assessment covers an AI-based customs classification tool currently in supervised training, with known bugs in its downstream calculation logic (duty, VAT, or exchange-rate computation), covering what typically works in such tools, common failure points and my recommended improvement and QA plan.

What Typically Works

Text-to-HS-code classification tools generally perform well when scoped narrowly and given specific, well-formed product descriptions:

- **Reference-matched classification:** Embedding-based similarity search against a curated HS code and description reference set, paired with a classifier fine-tuned on historical customs declarations, typically achieves strong accuracy on common, high-volume goods (textiles, electronics, vehicles) where labelled history is abundant.
- **Confidence-gated suggestions:** Presenting a ranked top-3 suggestion list with confidence percentages, rather than forcing a single answer, is the pattern that works best in production it keeps a human customs officer in the loop for anything uncertain instead of the tool silently guessing.
- **Narrow scope:** Scoping the tool to one country's tariff schedule (rather than the full global nomenclature) meaningfully improves accuracy, since HS interpretation and preferential-agreement rules are jurisdiction-specific.

Common Failure Points

- **Ambiguous product descriptions :** Vague descriptions such as “spare parts” or “machinery accessories” give the model too little to work with; it tends to confidently pick a wrong chapter rather than flag uncertainty.
- **Hierarchy confusion :** HS codes are hierarchical (chapter → heading → subheading). Models often get the broad chapter right but the fine-grained subheading wrong, precisely where duty rates diverge.
- **Data drift:** Tariff schedules and preferential rates (e.g. SADC/COMESA/AfCFTA) are revised periodically. A model trained on last year's data silently applies stale rules unless reference tables are versioned and kept in sync.
- **Calculation bugs downstream of classification:** This is usually a systems bug, not a model bug: wrong rate-lookup keys, currency/exchange-rate mismatches, VAT applied on the wrong base (CIF vs. FOB), or rounding inconsistencies. These are the most damaging failures because they look like classification errors when the classification was actually correct.
- **Lack of explainability:** Without a “why this code” trace, customs officers cannot sanity-check or dispute a suggestion, which erodes trust and slows adoption.

Recommended Improvement and QA Plan

- **Separate concerns explicitly:** Audit classification accuracy and calculation accuracy as two independent test suites, don't let a duty-calculation bug get misdiagnosed as a model accuracy problem.
- **Golden test set:** Build a held-out set of real, adjudicated declarations (500+ ideally, spanning all major chapters); track per-chapter precision/recall and a confusion matrix on every retrain.
- **Confidence-gated human review:** Route anything below a confidence threshold (e.g. 85%) to manual review rather than auto-approving; log officer overrides as new training signal.
- **Unit and regression tests on the calculation engine:** Fixed input→output test cases for duty/VAT math, exchange-rate application, and preferential trade agreement logic, run in CI before every deploy, the fastest way to catch the calculation bugs described in this brief.
- **Versioned tariff/reference tables:** Treat HS code and duty rate tables as versioned data with an effective date range, so historical declarations remain re-computable against the rules that applied at the time.
- **Shadow-mode rollout:** Run the retrained or fixed model alongside the current process for 2–4 weeks, comparing outputs before it is allowed to auto-approve anything, to catch regressions before they reach production.

Conclusion

The classification component of this tool is likely closer to production-ready than the calculation layer wrapped around it. The recommended path is to decouple the two, validate each independently against a golden test set and a CI-enforced calculation test suite and only promote the tool out of supervised training once both pass a shadow-mode comparison period against current manual output.